Explain what the forecast means and what limits it

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Revision / ES: revisión / PT: revisão

A corrected step, shown — and how it changes later results.

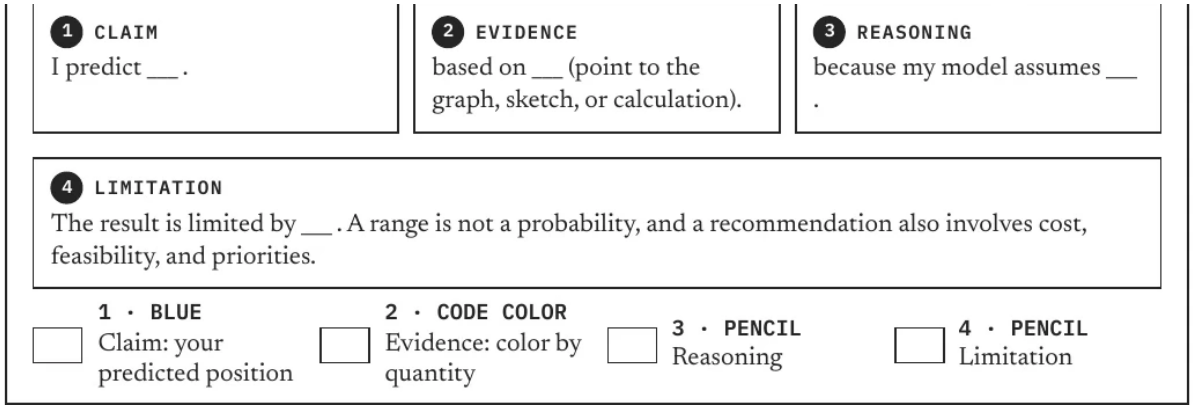
Also on your note page: Claim, Evidence, Limitation.

B READ THE RELATIONSHIP

A revision changes a forecast when new evidence or a corrected step calls for it.

Report the forecast, its bounds, and its limitations

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Defend a prediction: a 2 kg cart under 4 N net force, from rest at 1 m, for 2 s; model bound ± 1 m.

Given F net = 4 N, m = 2 kg x i = 1 m, t = 2 s bound ± 1 m

Rule force → a → v → x

Work a = 2 m/s², v = 4 m/s Δx = 4 m, x f = 5 m

Answer 4 to 6 m assumes constant forces

E SAVE FOR THE FINAL MEETING

The desk wants a claim it can defend: where XJ will be, how sure you are, and what your model leaves out. That is what makes a prediction trustworthy.

Read the physics: Revision

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Give a predicted position, a time and a reference frame. Show the measurement and calculation that support your answer. List any assumptions.

02 HOW IT WORKS

A model may give a central position and a range. A bound says which values the model allows; it does not give the chance of an impact. Compare the old and new results on one scale.

03 WHAT TO CHECK

Name the step that changed. Explain what it did to the forecast and what the physics still cannot decide.

04 TRY THE EXAMPLE FROM YOUR NOTES

Defend a prediction: a 2 kg cart under 4 N net force, from rest at 1 m, for 2 s; model bound ± 1 m. 4 to 6 m assumes constant forces.

05 CONNECT TO XJ

The desk wants a claim it can defend: where XJ will be, how sure you are, and what your model leaves out. That is what makes a prediction trustworthy.

SAY IT BEFORE YOU WRITE

Which piece of evidence would make you revise again? We revised ___ after ___. Our forecast still assumes ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Claim

ES: afirmación / PT: afirmação

Meaning in my words	How can you tell?
How would you explain claim to a friend? In my words, this means ___.	Which part of your forecast tells the listener exactly what you expect? I can tell because ___.
Example or sketch	**Non-example and why**
Make a claim for a cart moving steadily east at 2 m/s from the origin. My example shows ___.	"Here are my three readings." What is still missing from your report? I would change ___ because ___.

Evidence

ES: evidencia / PT: evidência

Meaning in my words	How can you tell?
How would you explain evidence to a friend? In my words, this means ___.	What reading supports the claim that the cart reached 4 m in 2 s? I can tell because ___.
Example or sketch	**Non-example and why**
How could two measurements support a claim that one cart accelerated more? My example shows ___.	"I expected our cart to win." Does that tell us which cart crossed first? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Limitation

ES: limitación / PT: limitação

Meaning in my words	How can you tell?
How would you explain limitation to a friend? In my words, this means ___.	Why would two positions a minute apart tell you little about motion an hour later? I can tell because ___.
Example or sketch	**Non-example and why**
Name one condition that could make a straight-line forecast for XJ unreliable. My example shows ___.	"Our answer is bad." How could that become a useful limitation? I would change ___ because ___.

Revision

ES: revisión / PT: revisão

Meaning in my words	How can you tell?
How would you explain revision to a friend? In my words, this means ___.	Your predicted sighting is +6 km and the new image reads +5 km. What would you update? I can tell because ___.
Example or sketch	**Non-example and why**
What would you tell the next crew after that sighting? My example shows ___.	A team changes its answer only because another team disagrees. What would justify the change? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Doodle two forecast bands on the same scale. Add a note connecting the revision to a specific measurement.

Labels you can use: claim · evidence · limit.

Choose zero, direction, and scale.



Tell a partner

We revised ___ after ___. Our forecast still assumes ___.

Would you air this headline?

22

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare forecast ranges and keep conclusions within the data.

CHOOSE ONE JOB TOGETHER

FORECAST ENGINEER ☐

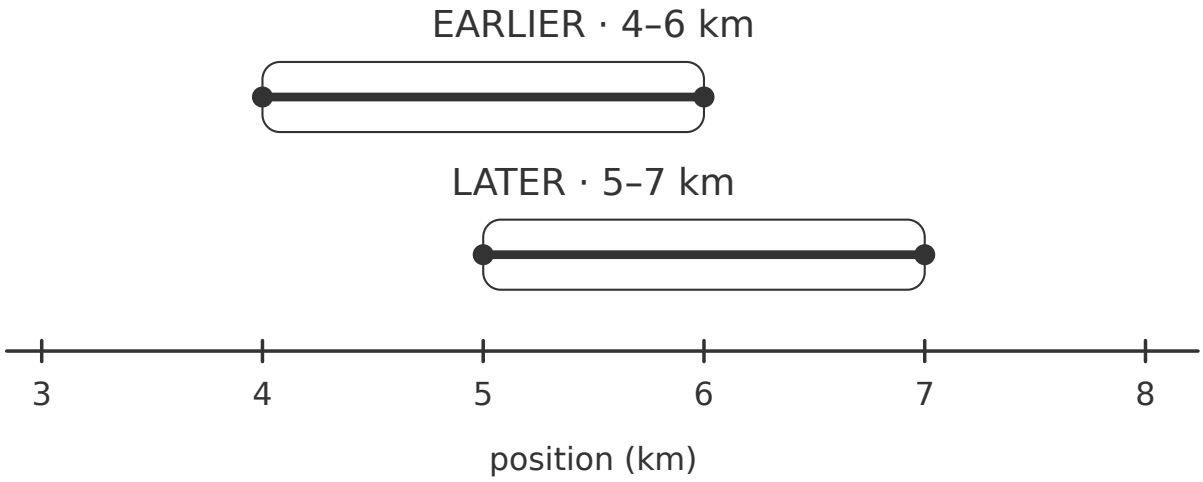
Mark both forecast bands. Issue a location bulletin and name an observation that would trigger another update.

SCIENCE NEWS EDITOR ☐

Decide whether to air the headline. Show both intervals and write the replacement line viewers should hear.

Word help: range / overlap / probability

Two short-interval forecasts in the same frame give **4-6 km** and **5-7 km**. The new estimate uses a later sighting. A headline says “Impact confirmed.”



01 OUR FIRST CALL

What location can you report? Should the proposed headline air? Explain.

The ranges show ___. They do not show ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Answer the caller using the bands. What can your desk report, and what is still unknown?

Now we would say ___ because ___.

Switch speakers. What do the bands actually tell us?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks